

PARTS LIST

Semiconductors:

IC1, IC8	- 7805 voltage regulator
IC2	- HT12E encoder
IC3	- 4011 quad 2-input NAND gate
IC4	- 74LS00 quad 2-input TTL NAND gate
IC5	- HT12D decoder
IC6, IC9	- NE555 timer
IC7	- 74LS164 8-bit shift register
T1, T2	- BC547 npn transistor
T3	- SL100 npn transistor
D1-D6	- 1N4007 rectifier diode
LED1-LED11	- 5mm LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

R1	- 680-kilo-ohm
R2, R4, R8-R11,	
R19, R20	- 10-kilo-ohm
R3, R18	- 2.2-kilo-ohm
R5, R7-R14,	
R16, R17, R21	- 1-kilo-ohm
R6	- 33-kilo-ohm
R15	- 220-kilo-ohm

Capacitors:

C1, C3	- 1000 μ F, 25V electrolytic
C2, C4	- 100 μ F, 16V electrolytic
C5, C6	- 10nF ceramic disk
C7	- 1 μ F, 16V electrolytic
C8	- 10 μ F, 16V electrolytic

Miscellaneous:

CON1, CON3	- 9-pin female connector
CON2, CON4	- 2-pin terminal connector
CON5, CON6	- 2-pin connector
TX1	- 433MHz RF transmitter module
RX1	- 433MHz RF receiver module
F1	- 5A fuse with holder
RL1	- 12V, 1 changeover relay
X1, X2	- 230V AC primary to 12V-0-12V, 500mA secondary transformer
	- 230V AC mains power supply
	- Heat-sinks for 7805

running status. When power is first turned on, IC7 is reset by power-on reset circuit comprising R18 and C8.

As a result, all output bits of IC7 become logic low. Complement of the final register output Q7 (which is at logic low now) is fed back as input data to pins 1 and 2 of shift register IC7 through output pin 4 of IC3. So, with clock pulses coming from IC6, the counter advances and its outputs follow sequentially the logic levels tabulated in Table 1, first in forward and then in reverse order. The counter goes on repeating the sequence as long as the clock feeds the counter.

Now, eight outputs of IC7 are tied to eight address lines (A0 through A7) of decoder HT12D. So, at a particular time, when IC7 output matches with the address of the encoder, the decoder is enabled to receive data from pin 11 of IC3, and VT output (pin17) of IC5 goes to logic high. LED2 indicates the

LEDs Indications

Table 2

LEDs	Status	Purpose	Remark
LED1	on	Motor on	LED1 off means motor is off
LED2	blink	When a valid signal is received	When it does not blink it means signal is not received as water level increases
LED3	Off	Maximum water level is at sensor S7	LED3 on means tank is empty up to sensor S7
LED4	Off	Water level is at sensor S6	LED4 on means tank is empty up to sensor S6
LED5	Off	Water level is at sensor S5	LED5 on means tank is empty up to sensor S5
LED6	Off	Water level is at sensor S4	LED6 on means tank is empty up to sensor S4
LED7	Off	Water level is at sensor S3	LED7 on means tank is empty up to sensor S3
LED8	Off	Water level is at sensor S2	LED8 on means tank is empty up to sensor S2
LED9	Off	Water level is at sensor S1	LED9 on means tank is empty up to sensor S1
LED10	Off	Minimum water level at sensor S0	LED10 on means tank is empty up to sensor S0

Status of LEDs

Table 3

LED3	LED4	LED5	LED6	LED7	LED8	LED9	LED10
ON(1)	ON(1)	ON(1)	ON(1)	ON(1)	OFF(0)	OFF(0)	OFF(0)

Status of Sensors

Table 4

S7	S6	S5	S4	S3	S2	S1	S0
EMPTY(1)	EMPTY(1)	EMPTY(1)	EMPTY(1)	EMPTY(1)	FULL(0)	FULL(0)	FULL(0)

reception of a valid signal (VT). This VT signal is complimented by gate N3 of IC3 and is fed to reset pin 4 of clock generator IC6. Hence, as soon as the decoder (IC5) is enabled to receive data, clock reset pin goes to logic low, barring its pulse generation.

The counter stops, as it is not getting any clock pulse from IC6. LED2 remains on, while LED11 goes off to indicate this situation. LEDs connected with the address lines indicate valid addresses at this instant, which is again the status of the sensors at the encoder. Glowing of LEDs (LED3 through LED10) represents logic high, which means the tank is empty (no water), and vice-versa (refer table). For example, status of the LEDs is shown in Table 3.

Water level in the corresponding sensor outputs is shown in Table 4.

Level of water in the tank will be above S2, but below S3.

Relay control

Water level in the tank can be confined between any two predetermined sensor levels. In the circuit diagram, S1 and S6 are chosen for minimum and maximum water levels, respectively. Whenever water level goes

below S1, 8-bit sensor output becomes as given in Table 5.

Water Level Below S1

Table 5

S7	S6	S5	S4	S3	S2	S1	S0
1	1	1	1	1	1	1	0

Address scanning at the receiver stops and, consequently, VT goes high when the Johnson counter 74LS164 outputs (Q0 through Q7) becomes as shown in Table 6.

Counter Output When S0 = 0

Table 6

Q7	Q6	Q5	Q4	Q3	Q2	Q1	Q0
1	1	1	1	1	1	1	0

Input pins 4 and 5 of NAND gate N2 of IC4 are tied to VT (pin 17) of IC5 and Q1 (pin 4) of IC7, respectively. As NAND gate N2 of IC4 gets two logic high signals at its pins 4 and 5, its output pin 6 goes to logic low, triggering the bi-stable flip-flop constructed around NE555 (IC9). The IC9 output (pin 3) is set to go high, turning on T2 and T3 (Darlington pair). The relay immediately gets activated and closes the contact. The pump motor is turned on, and stays on as long as the flip-flop is not reset.